torch.nn.utils.clip_grad_norm_#

https://pytorch.org/docs/stable/generated/torch.nn.utils.clip_grad_norm_.htm

Description:

Clip the gradient norm of an iterable of parameters. The norm is computed over the norms of the individual gradients of all parameters, as if the norms of the individual gradients were concatenated into a single vector. Gradients are modified in-place. This function is equivalent to torch.nn.utils.get_total_norm() followed by torch.nn.utils.clip_grads_with_norm_() with the total_norm returned by get_total_norm. parameters (Iterable[Tensor] or Tensor) – an iterable of Tensors or a single Tensor that will have gradients normalized max_norm (float) – max norm of the gradients

Function Signature

```
torch.nn.utils.clip_grad_norm_(parameters, max_norm,
norm_type=2.0, error_if_nonfinite=False, foreach=None)
[source]#

Parameters

Returns

Return type
```

Returns:

Tensor

Detailed Documentation

Documentation

Clip the gradient norm of an iterable of parameters.

The norm is computed over the norms of the individual gradients of all parameters, as if the norms of the individual gradients were concatenated into a single vector. Gradients are modified in-place.

This function is equivalent to torch.nn.utils.get_total_norm() followed by torch.nn.utils.clip_grads_with_norm_() with the total_norm returned by get_total_norm.

parameters (Iterable[Tensor] or Tensor) – an iterable of Tensors or a single Tensor that will have gradients normalized

max_norm (float) – max norm of the gradients

norm_type (float, optional) – type of the used p-norm. Can be 'inf' for infinity norm. Default: 2.0

error_if_nonfinite (bool, optional) – if True, an error is thrown if the total norm of the gradients from parameters is nan, inf, or -inf. Default: False

foreach (bool, optional) – use the faster foreach-based implementation. If None, use the foreach implementation for CUDA and CPU native tensors and silently fall back to the slow implementation for other device types. Default: None

Total norm of the parameter gradients (viewed as a single vector).